GYAN KUMAR SAH

Physics Researcher (Quantum & Condensed Matter) • Nepal / India • He/Him
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PROFILE

Final-year Physics Honours (with Research) student ranked at 9.69/10 CGPA, working at the intersection of experimental condensed-matter physics and quantum computing. Selected in the **top 100 of 65,000+** applicants nationally for the WISER quantum algorithms program. Two years of hands-on research and paid R&D engineering, with a track record of turning dense technical material — Shor's algorithm, BB84 key distribution, anomalous Hall measurements — into written explanations and working prototypes. Comfortable evaluating and correcting technical reasoning, and already building with LLM APIs. Fluent in English, Nepali, and Hindi.

EDUCATION

SRM University, Andhra Pradesh, India 2023 – 2027 (expected)

B.Sc. Physics, Honours with Research | CGPA 9.69 / 10.00 | 80% merit scholarship

- Coursework: Quantum Mechanics, Mathematical Methods in Quantum Computation, Statistical Physics, Solid State Physics, Nuclear & Particle Physics, Special Relativity, Cryptography, Elementary Number Theory, Atomic & Molecular Physics, Analog & Digital Electronics, Mathematical Modeling of Physical Data, Foundations of Data Analytics, R and Python, LabVIEW.

Global School of Science, Kathmandu, Nepal 2021 – 2023

Higher Secondary, Science Stream | GPA 3.81 / 4.00 | Merit scholarship

RESEARCH EXPERIENCE

Undergraduate Researcher — Exotic Interface Lab, Dept. of Physics, SRM AP 2024 – Present

Advisor: Dr. Krishna Prasad Maity | Funded by a competitive UG/PG SEED Research Grant

- **Optically Controlled Ferromagnetic Thin Films.** Deposited 30 nm cobalt in a masked Hall geometry by RF sputtering onto 5×5 mm glass substrates; measured the anomalous Hall effect with a Zurich Instruments MFLI lock-in amplifier.
- **Cryogenic Measurement System.** Built a low-temperature device-physics setup end to end — CNC-machined aluminium connectors and precision gasket sealing to hold 10^{-3} mbar, then wrote the LabVIEW program interfacing a Keithley SMU and a DC2200 LED driver to automate the full measurement sweep.
- **Photoresponse in CdSe / Graphene-Quantum-Dot Hybrids** (with N. George, A. Sampath). Drop-cast CdSe and GQD composites on silicon; ran intensity- and wavelength-resolved I–V measurements (Keithley 2636B, Thorlabs DC2200) under LabVIEW automation, plus lock-in time-response measurements. Cleaned and analysed the resulting datasets in Python and produced publication-quality figures in Origin.

INDUSTRY EXPERIENCE

Research Engineer Intern (paid) — Shakti Photon Solutions Pvt. Ltd. Feb 2025 – Present

Green hydrogen electrolysers, fuel cells, and CO_2 reduction | Guntur, Andhra Pradesh, India

- Designed components in Fusion 360 and manufactured them by CNC and 3D printing across steel, graphite, acrylic, and PTFE; fabricated membrane electrode assemblies by ultrasonic spray coating and hot pressing.
- Sole owner of an Arduino + MQ-8 hydrogen leak detection system, including the custom 3D-printed housing.
- Produced the 30×30 cm electrolyser design for a large industrial order.
- Delivered and commissioned the fuel-cell balance of plant at **IIT Delhi** and trained their technical staff on operation — a teaching-under-scrutiny experience that fed directly into the cryogenic build above.
- Designed and shipped the company's corporate website (Vite, deployed on Vercel with SEO and sitemap).

QUANTUM COMPUTING, CRYPTOGRAPHY & AI PROJECTS

- **WISER Quantum Algorithms & Hands-on Training** (Dec 2025 – Jun 2026). National two-phase program run with Amaravati Quantum Valley and Qubitech. Advanced from 65,000+ enrolled learners, to the top 3,000,

through a proctored in-person examination, to the **final 100 awarded a 100% scholarship**; taught by MIT and Princeton faculty and quantum-startup founders.
- **Quantum Double Shield Protocol** — 3rd place, Amaravati Quantum Valley Hackathon (Aug 2025). Two-layer post-quantum communication scheme combining BB84 quantum key distribution with a quantum random number generator for adaptive authentication; simulated in Qiskit to demonstrate secure key exchange and eavesdropper detection. qdsp.lovable.app
- **Post-Quantum Cryptography: From RSA to Shor's Algorithm** (Nov 2024, Dr. Sazzad Ali Biswas). Written study of why RSA's hardness assumption collapses under Shor's algorithm, and of the PQC schemes proposed to replace it. Full paper
- **Email-to-Calendar AI Agent** (May 2026). Autonomous agent in Google Apps Script calling the Mistral AI API to extract structured event details (time, link, attendees) from inbox mail into a relational store, with a zero-code AppSheet mobile front end for swipe-to-approve sync to Google Calendar.

Self-Initiated Software

- **Nepali Village Calculator** — free compound-interest calculator using Bikram Sambat dates, built for how village lending actually works in Nepal after watching my mother phone a neighbour to do the maths. Now the **#1 Google result** for "village calculator nepal". village-calculator.vercel.app
- **Portfolio Insight** — investment tracker for the Nepali market (NEPSE) covering real returns, bonus-share accounting, and sector-allocation dashboards. React, TypeScript, Vite, Tailwind, Recharts. github.com/sahgyan9/NEPSE-Portfolio-Insight
- **Friendly Learning SRMAP** — peer-to-peer mentoring and study-partner platform for my university. friendly-learning-srmap.lovable.app
- **Automatic Temperature-Controlled Fan** — BC107 transistor, 10 kΩ thermistor circuit with a 3D-printed enclosure; written up as a full report.

Skills

- **Physics & Quantum:** quantum mechanics, quantum algorithms (Shor, BB84, QKD), Qiskit, PennyLane, solid-state and condensed-matter physics, statistical physics, number theory and cryptography.
- **Experimental:** RF/DC sputtering, lock-in amplification, four-probe and GI-XRD, Keithley SMU, cryogenics and vacuum systems, instrument calibration, experimental design.
- **Computational:** Python (NumPy, SciPy, Matplotlib, pandas), R, LabVIEW automation, Origin, data cleaning and analysis, LaTeX, Git, JavaScript/TypeScript, React.
- **Communication:** technical writing and full research reports, teaching and staff training, documentation, public speaking. English (fluent), Nepali (native), Hindi (fluent).

Honours & Awards

- Top 100 of 65,000+ nationally — WISER Quantum Program, 100% scholarship (2026).
- 3rd place — Amaravati Quantum Valley Hackathon (2025).
- UG/PG SEED Research Grant, SRM University AP.
- 80% merit scholarship (SRM University AP); merit scholarship (Global School of Science).
- Featured in an SRM University AP student interview. Watch

Certifications

- AI Foundations — OpenAI (Jun 2026) • Quantum Computing and Programming — QWorld (Jun 2026)
- Quantiverse 2026: Hands-on Quantum Computing — C-DAC Noida (Feb 2026) • Descriptors & Machine Learning for Materials — SRM AP (Feb 2026)
- Data Science with Python — Finlatics (Aug 2025) • The Art of Doing Electronics — Coursera (Nov 2025) • Quantum Computing Fundamentals — LinkedIn Learning (Jul 2024)

Leadership

- **Lead, Aadigyan Winter Youth Program** — International Relations Council, SRM AP (Feb 2024). Selected to the Council management team by written SOP and interview; ran a four-day cross-border program hosting 35+ students and teachers from Nepal, owning planning through execution.
- **School Captain** (2020) and **Discipline In-charge** (2019), Lord Bright Wisdom Int'l School. Highest student office; directed 33+ sub-captains across a student body of 1,400+, reporting to the Principal.